

Pension Run

Kai-Jyun Wang

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 - has worked for 30 years,
 - retiring at 65,
 - and lives 12 years after retirement.
- Lumpsum: 2,061,000 NTD.
- Monthly: 2,724,504 NTD. (2% annual rate)

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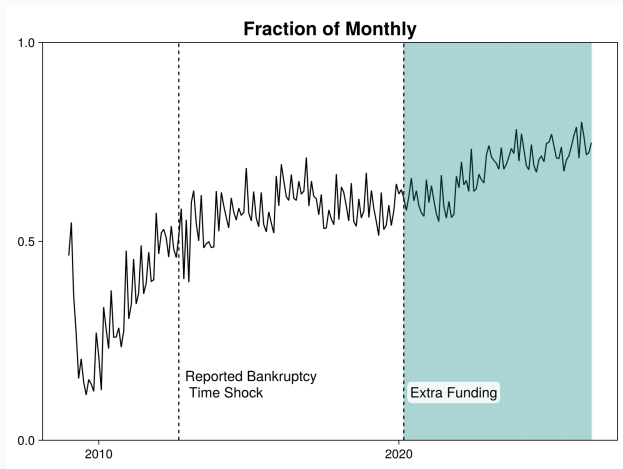
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- **This paper:** Insolvency is endogenous—annuitization choices change the fund's liabilities and its probability of depletion.

How many people are choosing monthly?



► [View data explanation](#)

Model

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- Benefits stop when the fund balance reaches zero.
- Retirees' choices affect fund outflows and, consequently, the value of the monthly pension.

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 - Stable normalized population density.
- In steady state, normalized population density $\hat{g} = g/N_t$, $N_t = \int g(t, s)ds$ satisfies

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m, \\ be^{-ns}e^{-m(s-T_m)}, & s \geq T_m, \end{cases}$$

where $dN_t = nN_t dt$ is the population growth rate.

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- The PDV of monthly is

$$V(\hat{B}, \hat{Q}) = \mathbb{E} \left[\int_0^{T_{\hat{B}}} e^{-(\rho+m)s} \underbrace{p}_{\text{monthly benefit}} ds \mid \hat{B}, \hat{Q} \right]$$

where $T_{\hat{B}}$ is the bankruptcy time.

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Detrended State Dynamics

- Define per capita states $\hat{B}_t = B_t/N_t$ and $\hat{Q}_t = Q_t/N_t$.

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- More lump-sum claims reduce the fund immediately.
- More monthly claims raise future liabilities through $\hat{Q}_t$.

Aggregation and Equilibrium

- Assume the idiosyncratic preference $\epsilon_{it}(d) \sim \text{Gumbel}(0, 1)$.

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 - q_t implies the state dynamics consistent with the belief on $T_{\hat{B}}$.

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- At a sufficiently large upper boundary, impose $\partial V / \partial \hat{B} \approx 0$.
- Solve by finite differences and successive iteration, equivalent to semi-implicit scheme with $dt = \infty$. [View algorithm](#)

Results

Calibration

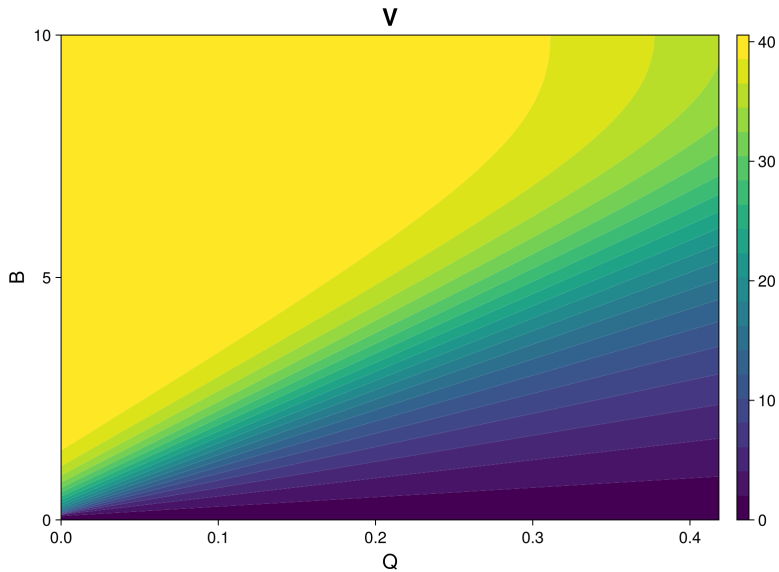
Parameter	Description	Value	Target
<i>Panel A: External Calibration</i>			
b	Birth rate	0.00462	Total fertility rate
m	Mortality rate	0.06	
T_w	Working age	25	
T_r	Retirement age	61	Average retirement age
T_m	Age of mortality exposure	60	
r	Pension fund return rate	0.093	Mean annualized pension fund return
τ	Pension tax	0.075	Pension tax
y	Income	5.268	Highest monthly insured salary
ℓ	Lump-sum pension	19.75	Transfer based on 30 years of working
p	Monthly pension	2.449	Pension based on 30 years of working

Note: The monetary unit in the model is 100,000 NTD.

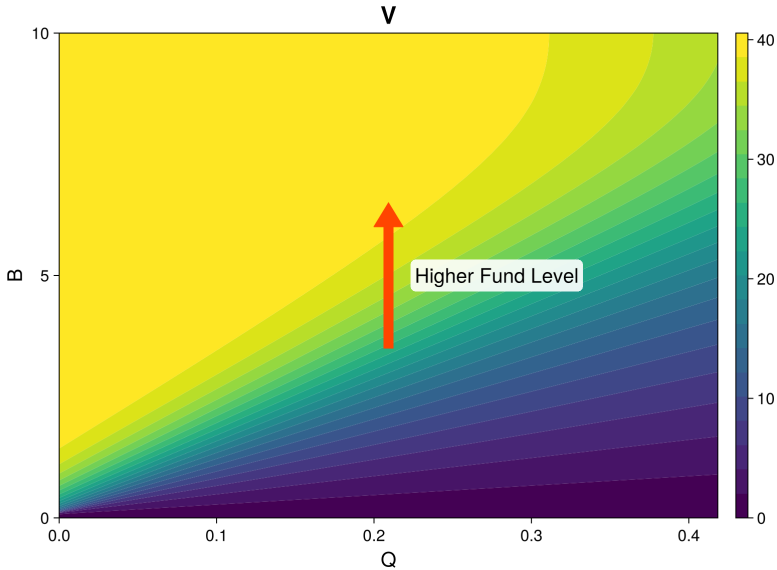
Parameter	Description	Value	Target
<i>Panel B: Internal Calibration</i>			
σ	Pension fund volatility	0.1044	$SD\left(\frac{\hat{B}_{t+\Delta} - \hat{B}_t - \mu_B(\hat{B}_t, \hat{Q}_t, \theta)\Delta}{\hat{B}_t\sqrt{\Delta}}\right) = \sigma$
α	Preference for lump-sum scheme	-1.334	$E\left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2\right]$
ρ	Subjective discount rate	$1.813e - 37$	$E\left[(q - q(\hat{B}_t, \hat{Q}_t, \theta))^2\right]$

Note: The monetary unit in the model is 100,000 NTD.

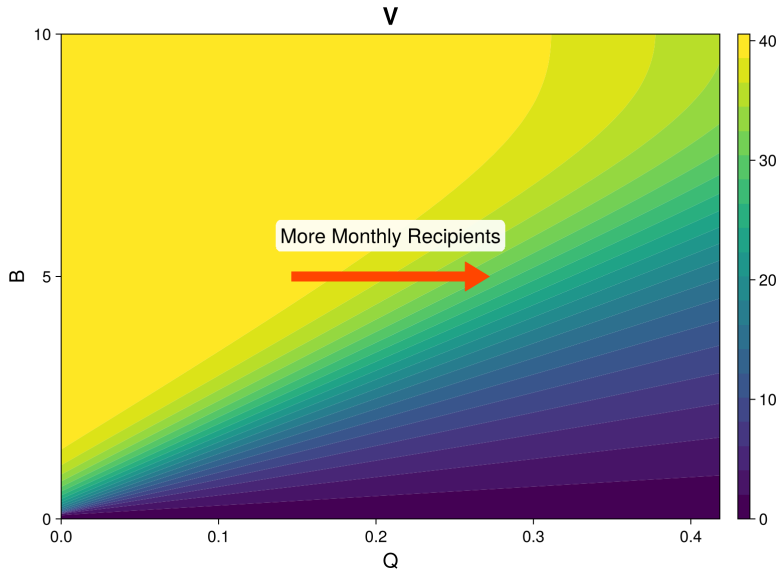
Estimated V



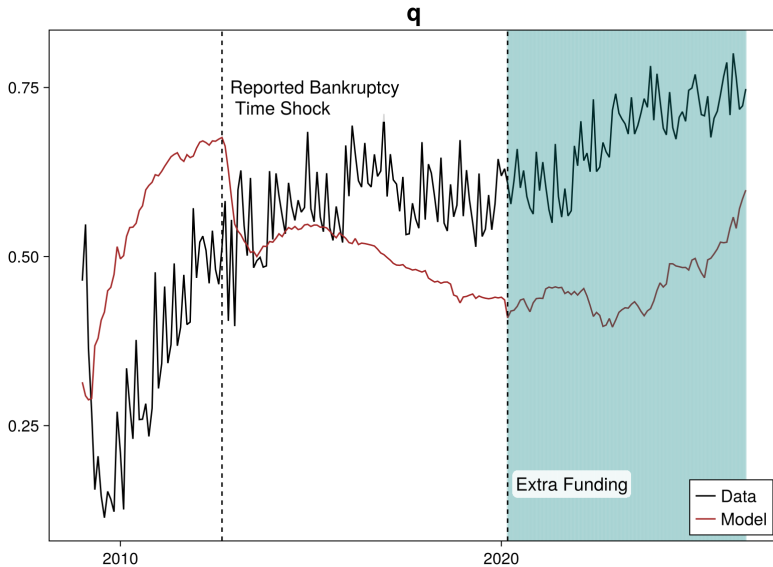
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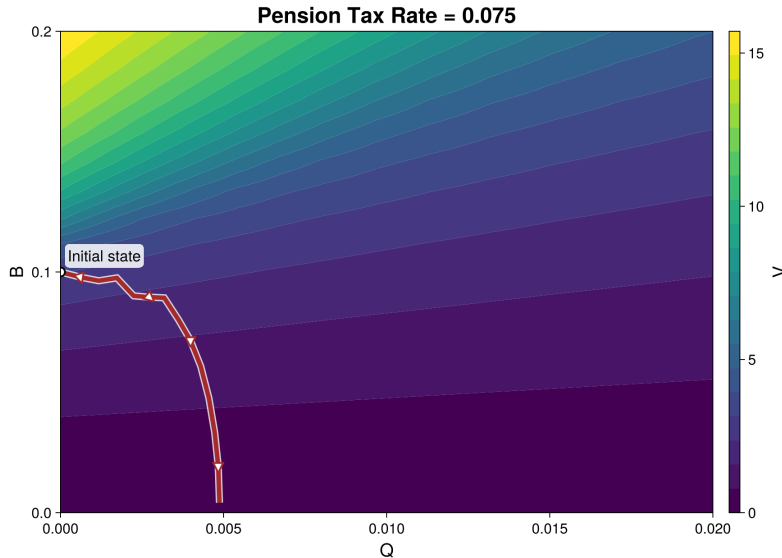
Model Fit



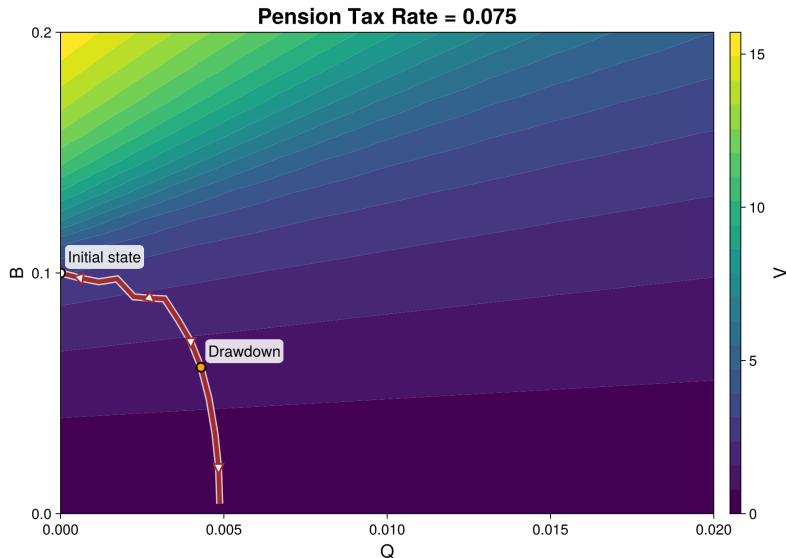
Low Pension Tax Rate



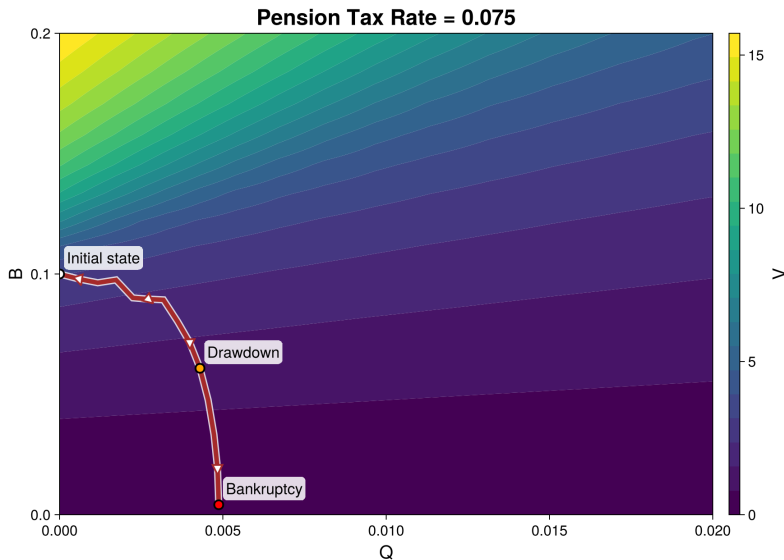
Low Pension Tax Rate



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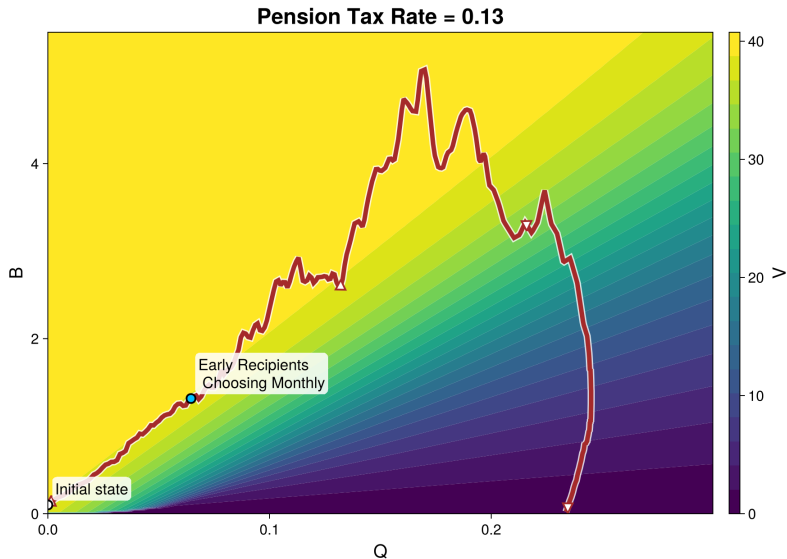
Medium Pension Tax Rate



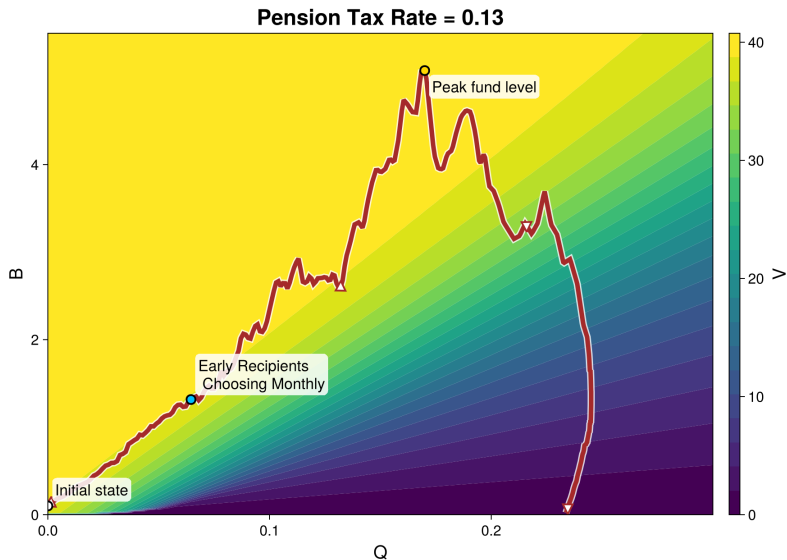
Medium Pension Tax Rate



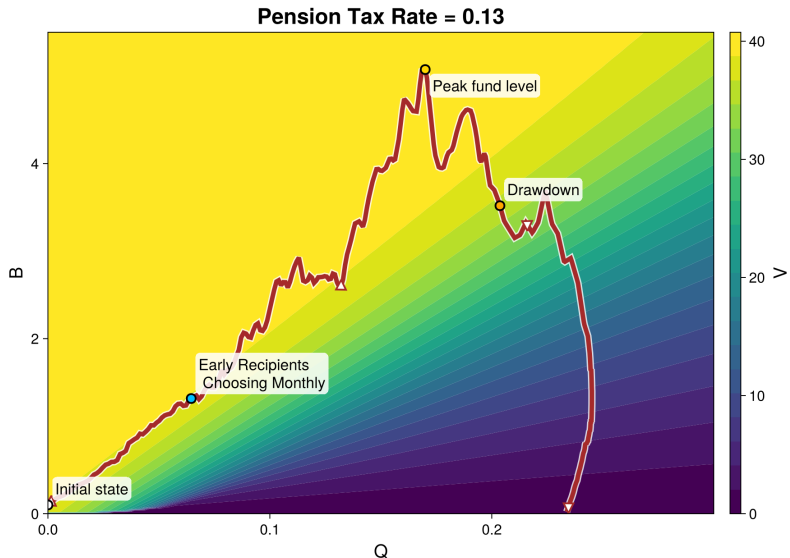
Medium Pension Tax Rate



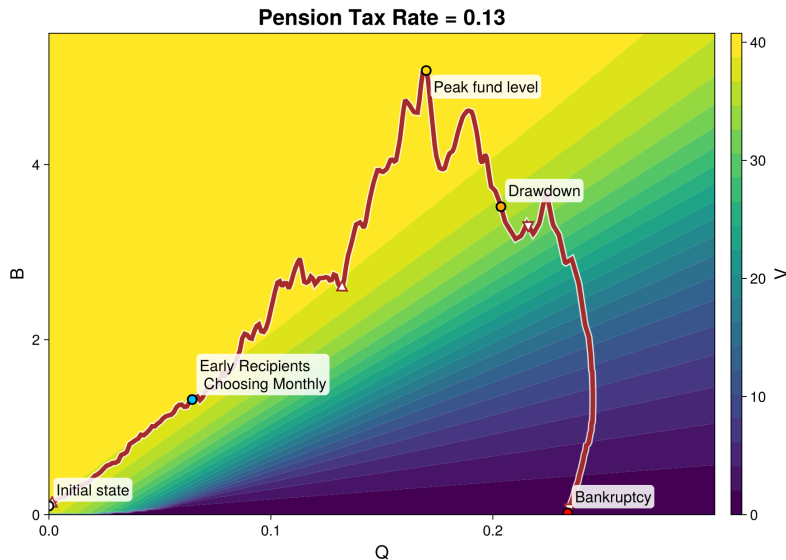
Medium Pension Tax Rate



Medium Pension Tax Rate



Medium Pension Tax Rate



Wrap up

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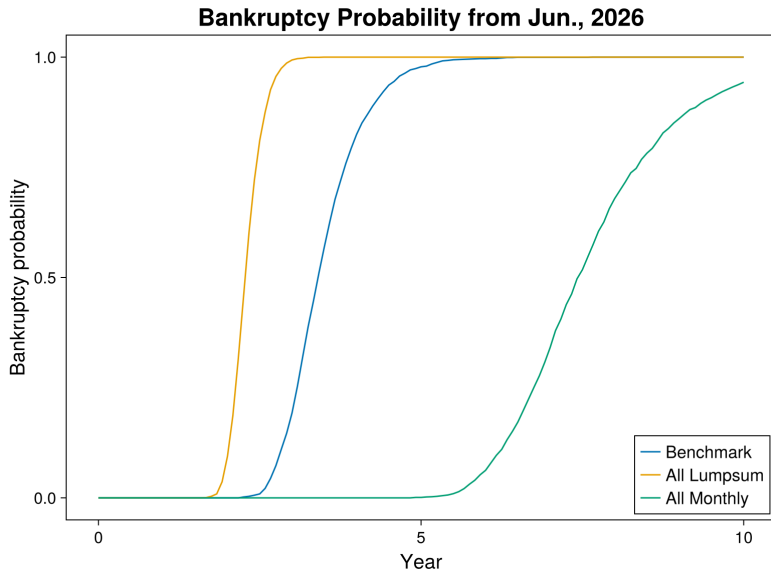
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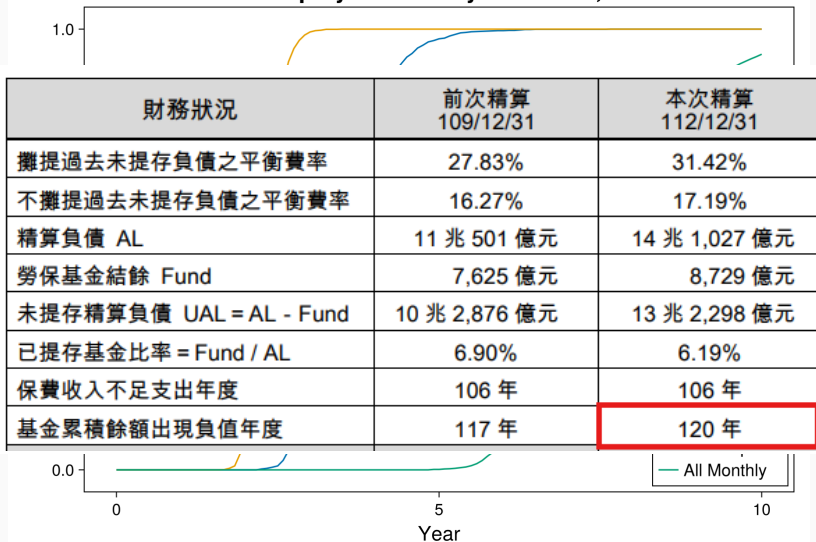
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Bankruptcy Probabilities

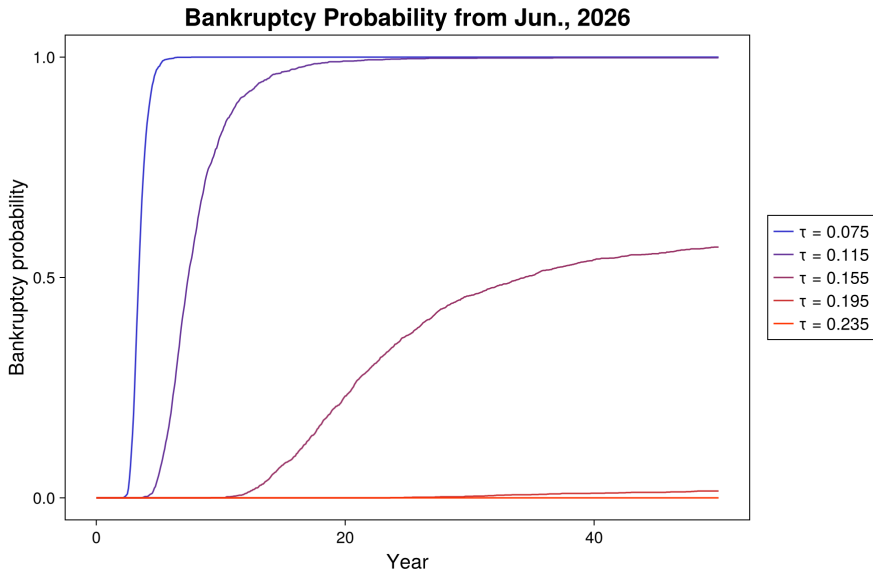


Bankruptcy Probabilities

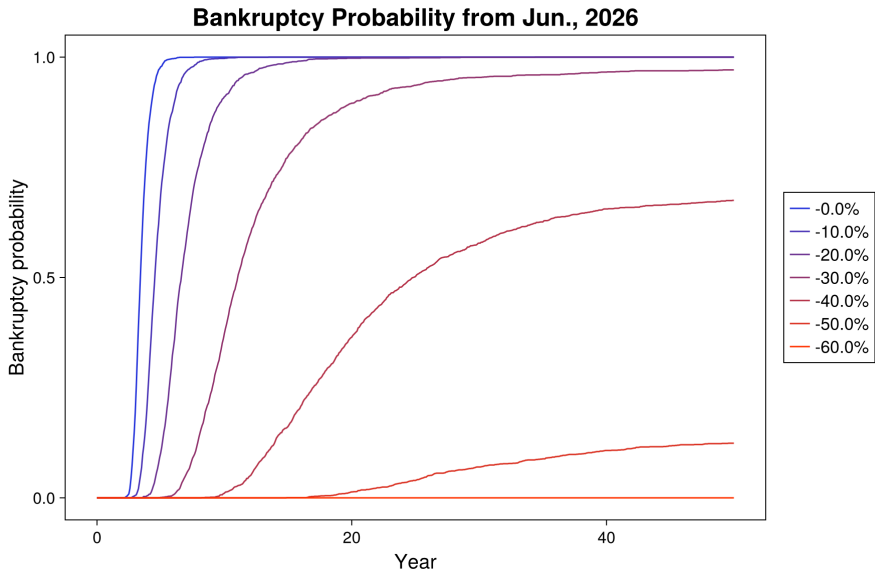
Bankruptcy Probability from Jun., 2026



Bankruptcy Probabilities for Different τ



Bankruptcy Probabilities for Cutting Benefits



Conclusions

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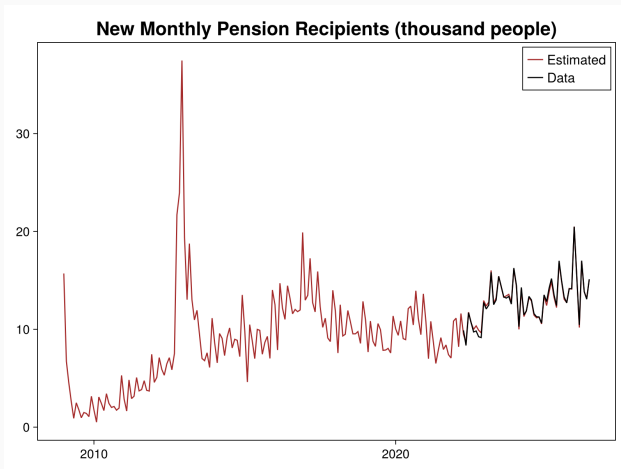
Appendix

- **Official monthly sources (since 2009):**
 - Bureau of Labor Funds: fund balance and investment return.
 - Bureau of Labor Insurance: stocks of monthly recipients and new lump-sum claims.
 - Ministry of the Interior: population and mortality at ages 60+.
- BLI reports new monthly recipients (first payments) only from May 2022. Before then, reconstruct entries from the stock of monthly recipients:

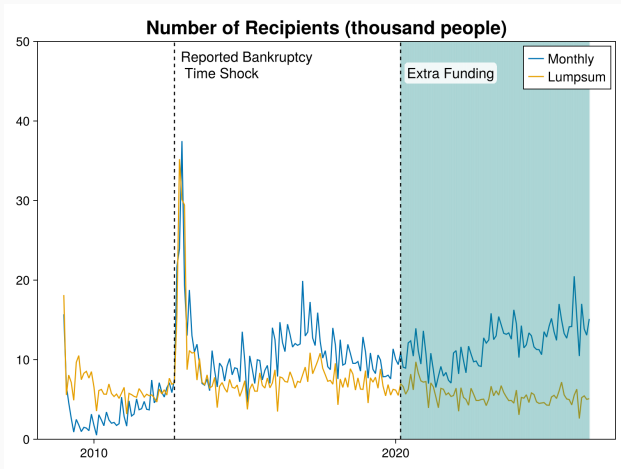
new monthly recipients = change in the recipient stock + estimated deaths
+ average correction from the overlap period.

Estimated deaths use the monthly mortality rate implied by annual mortality among Taiwan residents aged 60+; negative estimates are set to zero.

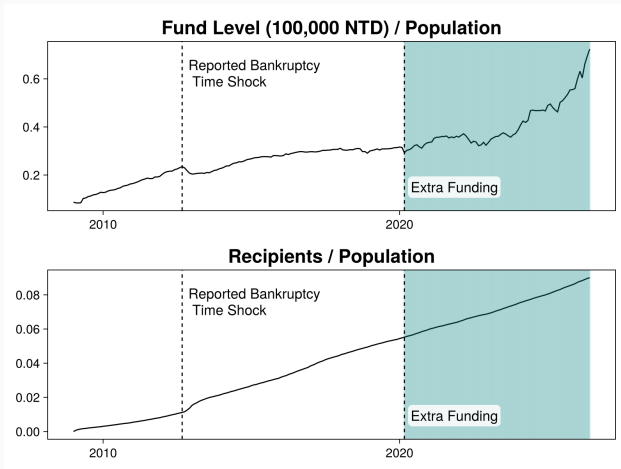
Reconstruction Validation



◀ Back to recipient fraction

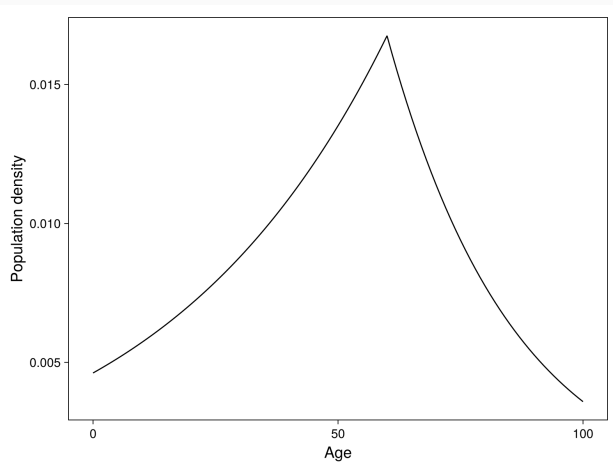


◀ Back to recipient fraction



[◀ Back to recipient fraction](#)

Population Density



[◀ Back to population](#)

Successive Iteration Algorithm

Require: Initial guess $V^{(0)}$, tolerance $\epsilon > 0$, maximum iterations K

1: Set $k \leftarrow 0$

2: **repeat**

3: Given $V^{(k)}$, compute

$$\mu_B^{(k)} = \mu_B(B, Q, V^{(k)}(B, Q)), \quad \mu_Q^{(k)} = \mu_Q(B, Q, V^{(k)}(B, Q))$$

and the infinitesimal generator

$$\mathcal{A}^{(k)} = \mu_B^{(k)} \frac{\partial}{\partial B} + \mu_Q^{(k)} \frac{\partial}{\partial Q} + \frac{1}{2} \sigma^2 B^2 \frac{\partial^2}{\partial B^2}.$$

with adjustments to the boundary nodes.

4: Solve the linear system

$$\left((\rho + m)I - \mathcal{A}^{(k)} \right) V^{(k+1)} = \mathbf{b}, \quad \mathbf{b} = \rho \text{vec} \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 \\ \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & 1 \end{pmatrix}.$$

5: Compute the error

$$e^{(k+1)} \leftarrow \left\| V^{(k+1)} - V^{(k)} \right\|_{\infty}.$$

6: $k \leftarrow k + 1$

7: **until** $e^{(k)} < \epsilon$ or $k = K$

8: **return** $V^{(k)}$.

◀ Back to numerical solution